PYTHON LIBRARY FILES IMPLEMENATATION

1. FILTERING DATA:

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df[df["score"]>15]
print(X)
```

```
        name  score  attempts qualify
2  Katherine   16.5         2     yes
5    Michael   20.0         3     yes
```

2. FILTERING DATA WITH MULTIPLE CONDITION:

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df[(df["score"]>15)&(df["attempts"]<3)]
print(X)
```

```
        name  score  attempts qualify
2  Katherine   16.5         2     yes
```

3. SORTING DATA(ASC):

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.sort_values("score")
print(X)
```

```
...
          name  score  attempts qualify
1         Dima    9.0         3      no
4        Emily    9.0         2      no
3        James   12.0         3      no
0    Anastasia   12.5         1     yes
7        Laura   13.5         1      no
6      Matthew   14.5         1     yes
2    Katherine   16.5         2     yes
5      Michael   20.0         3     yes
```

SORTING DATA(DESC):

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.sort_values("score",ascending = False)
print(X)
```

```
        name  score  attempts qualify
5    Michael   20.0         3     yes
2  Katherine   16.5         2     yes
6    Matthew   14.5         1     yes
7      Laura   13.5         1      no
0  Anastasia   12.5         1     yes
3      James   12.0         3      no
1       Dima    9.0         3      no
4      Emily    9.0         2      no
```

4. HANDLING MISSING VALUES(DETECTING) :

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.isnull()
print(X)
```

```
    name  score  attempts  qualify
0  False  False     False    False
1  False  False     False    False
2  False  False     False    False
3  False  False     False    False
4  False  False     False    False
5  False  False     False    False
6  False  False     False    False
7  False  False     False    False
```

HANDLING MISSING VALUES(COUNTING) :

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.isnull().sum()
print(X)
```

```
name        0
score       0
attempts    0
qualify     0
dtype: int64
```

5. REMOVING MISSING VALUES :

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.dropna()
print(X)
```

```
        name  score  attempts qualify
0  Anastasia   12.5         1     yes
1       Dima    9.0         3      no
2  Katherine   16.5         2     yes
3      James   12.0         3      no
4      Emily    9.0         2      no
5    Michael   20.0         3     yes
6    Matthew   14.5         1     yes
7      Laura   13.5         1      no
```

6. REMOVING DUPLICATION VALUES :

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
X= df.drop_duplicates()
print(X)
```

```
         name  score  attempts qualify
0  Anastasia   12.5         1     yes
1       Dima    9.0         3      no
2  Katherine   16.5         2     yes
3      James   12.0         3      no
4      Emily    9.0         2      no
5    Michael   20.0         3     yes
6    Matthew   14.5         1     yes
7      Laura   13.5         1      no
```

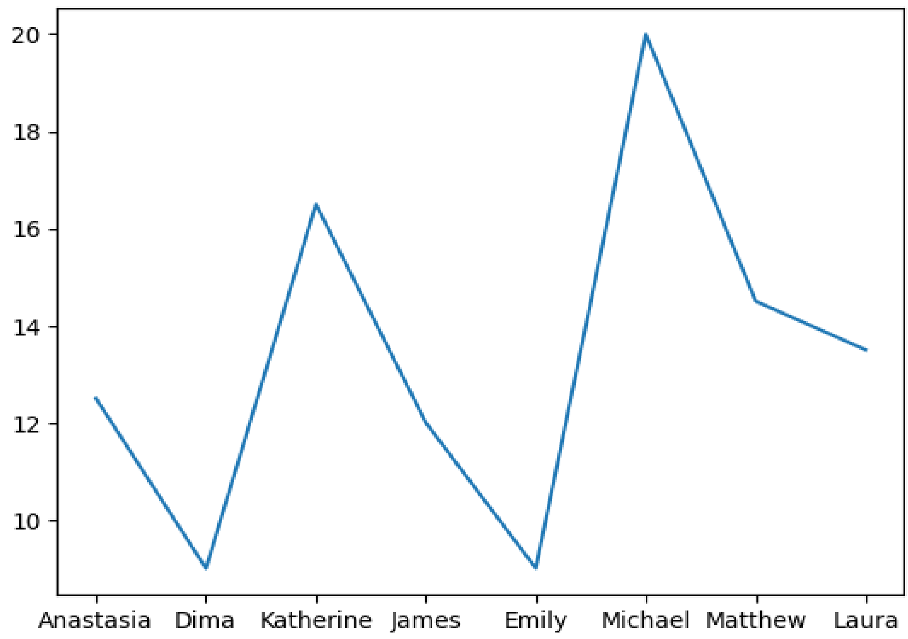
7. AGGREGATE FUNCTIONS:

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
print("Avg:",df["score"].mean())
print("highest mark:",df["score"].max())
print("lowest mark:",df["score"].min())
```

```
Avg: 13.375
highest mark: 20.0
lowest mark: 9.0
```

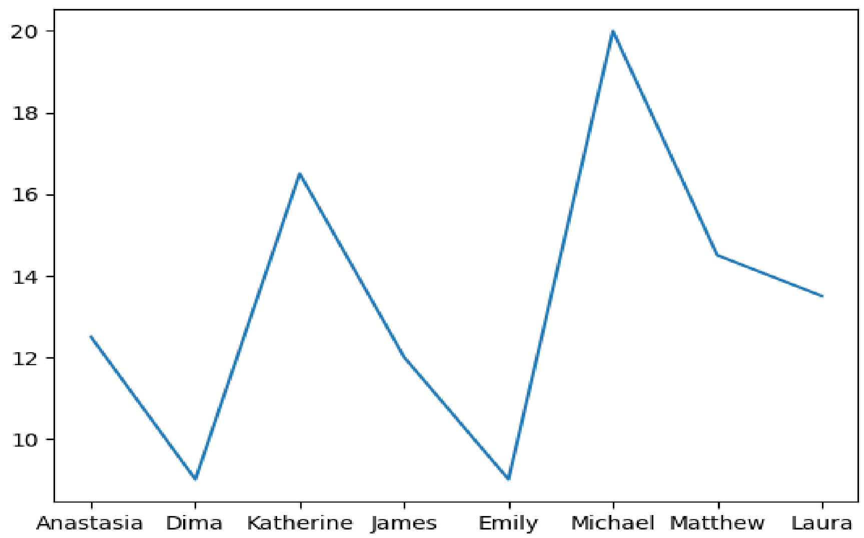
8. DATA VISUALIZATION:

```
import pandas as pd
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.plot(df["name"],df["score"])
plt.show()
```



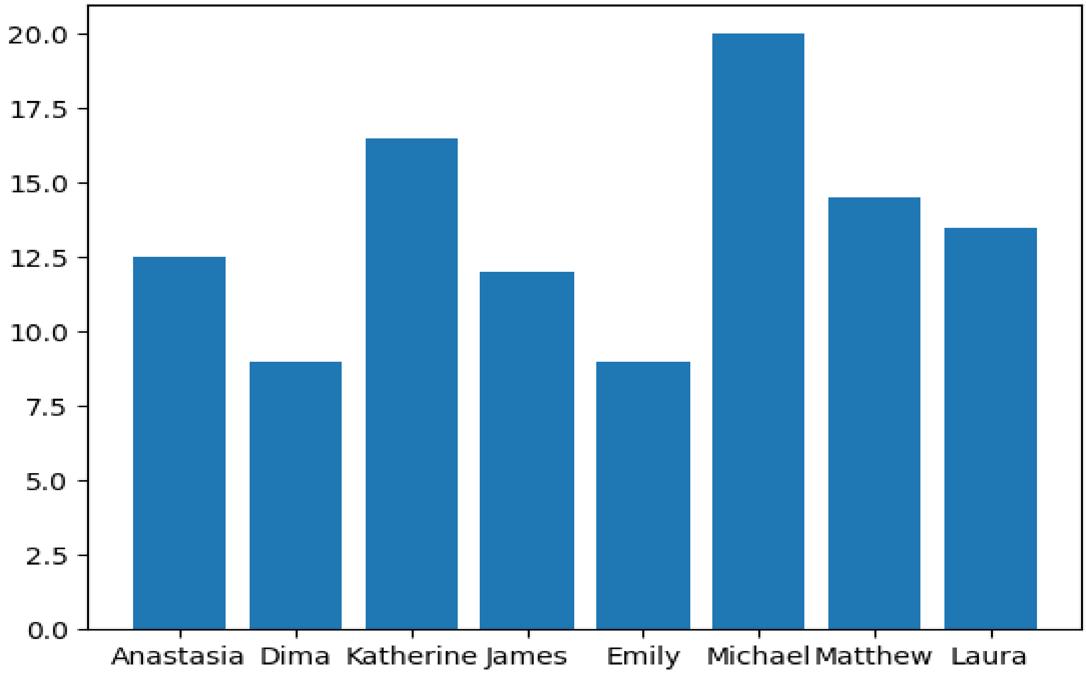
9. LINE PLOT:

```
import pandas as pd
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.plot(df["name"],df["score"])
plt.show()
```



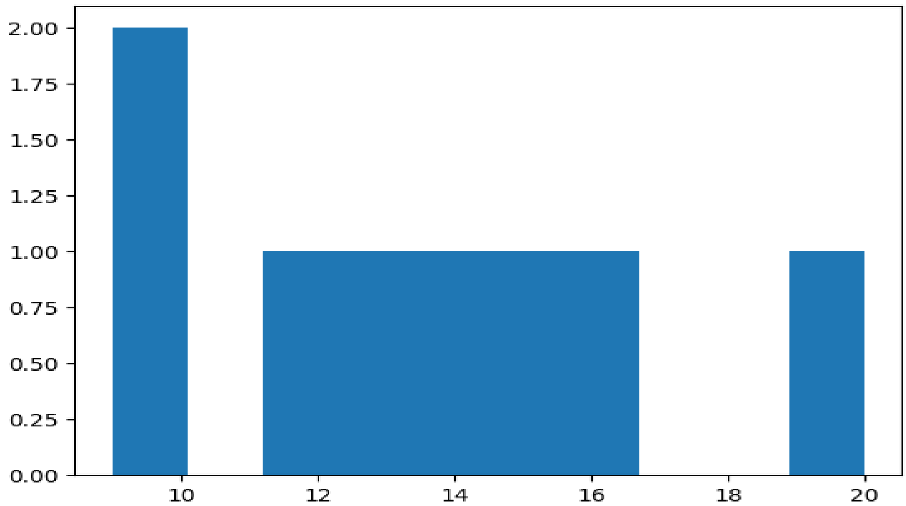
10. BAR CHART:

```
import pandas as pd
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.bar(df["name"],df["score"])
plt.show()
```



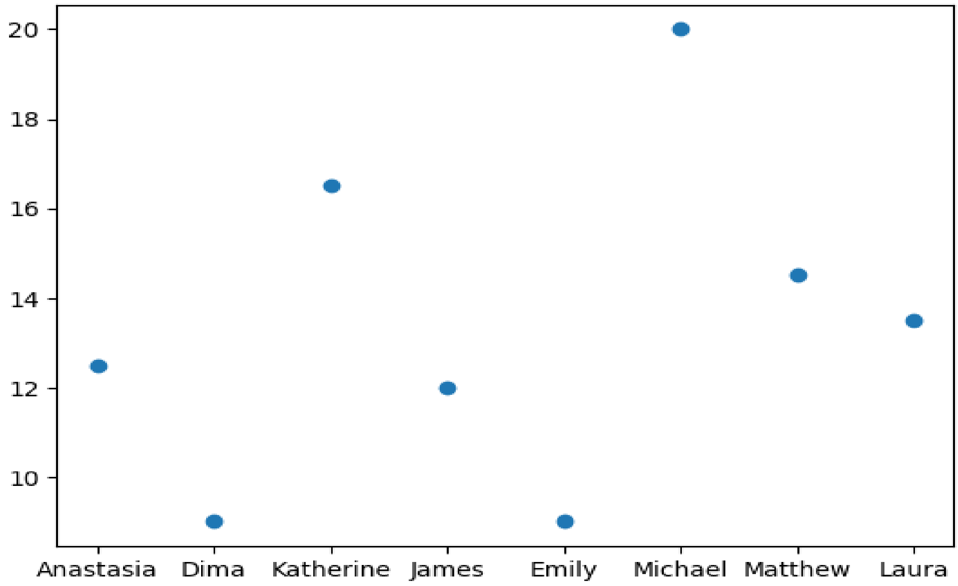
11. HISTOGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.hist(df["score"])
plt.show()
```



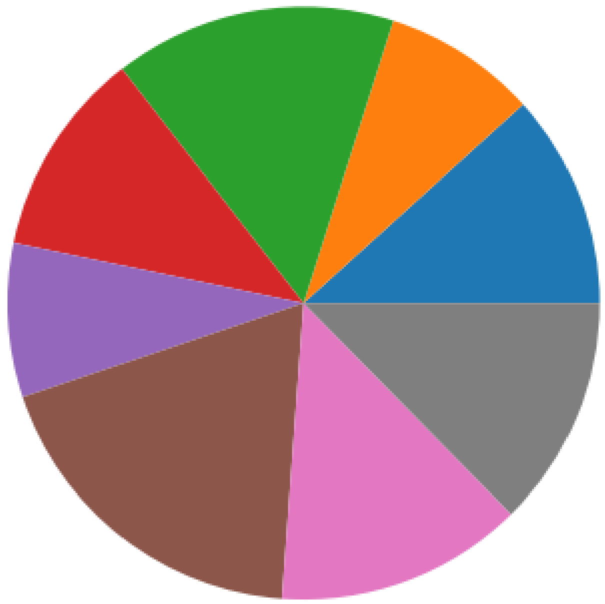
12. SCATTER PLOT:

```
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.scatter(df["name"],df["score"])
plt.show()
```



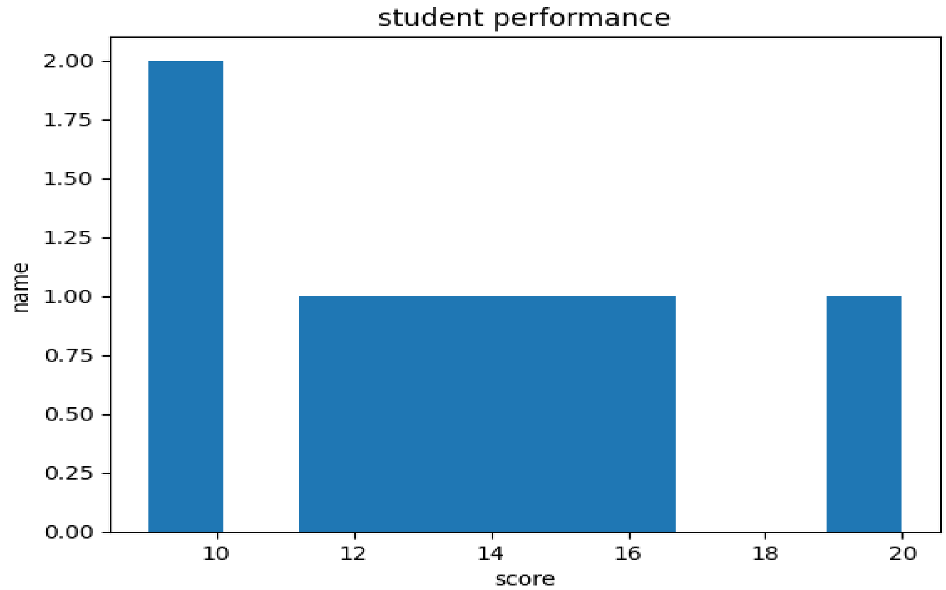
13. PIE CHART

```
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.pie(df["score"])
plt.show()
```



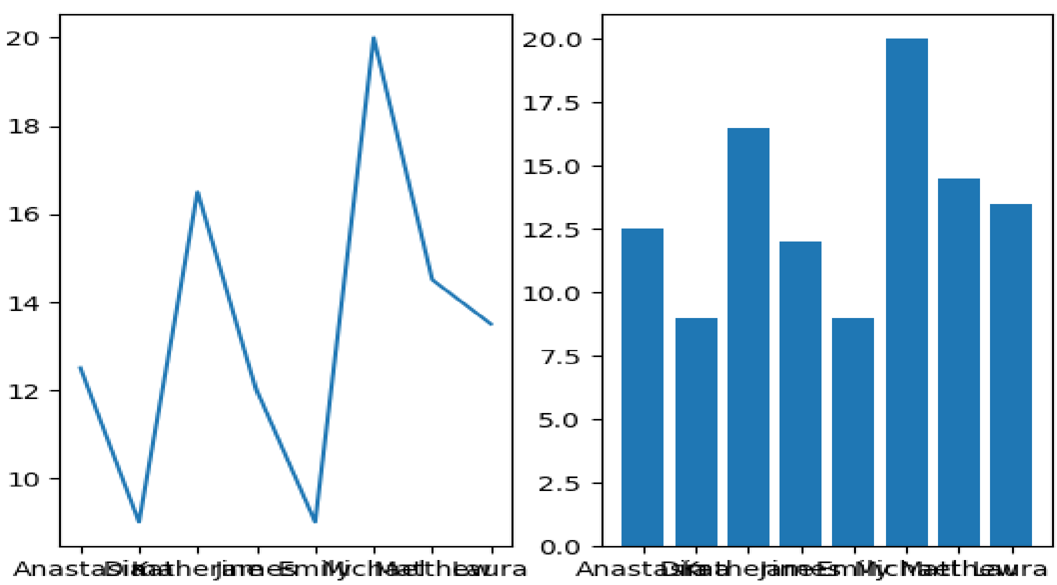
14. PLOT LABELS & TITLES

```
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.hist(df["score"])
plt.xlabel("score")
plt.ylabel("name")
plt.title("student performance")
plt.show()
```



15. SUB-PLOTS

```
import matplotlib.pyplot as plt
df= pd.read_csv("exam_data.csv")
plt.subplot(1,2,1)
plt.plot(df["name"],df["score"])
plt.subplot(1,2,2)
plt.bar(df["name"],df["score"])
plt.show()
```



16. READING CSV FILE:

```
import pandas as pd
df= pd.read_csv("exam_data.csv")
```

17. WRITING CSV FILE:

```
df.to_csv("new.csv")
```

IMPORTANT DATAFRAME FUNCTION:

1. head()

2. tail()

3. info()

4. describe()

```
[25] print(df.head())

         name  score  attempts qualify
0   Anastasia   12.5         1     yes
1        Dima    9.0         3      no
2   Katherine   16.5         2     yes
3       James   12.0         3      no
4       Emily    9.0         2      no
```

```
[26] print(df.tail())

       name  score  attempts qualify
3     James   12.0         3      no
4     Emily    9.0         2      no
5   Michael   20.0         3     yes
6   Matthew   14.5         1     yes
7     Laura   13.5         1      no
```

```
[27] print(df.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   name      8 non-null      object
 1   score     8 non-null      float64
 2   attempts  8 non-null      int64
 3   qualify   8 non-null      object
dtypes: float64(1), int64(1), object(2)
memory usage: 388.0+ bytes
None
```

```
[28] print(df.describe())

           score  attempts
count   8.000000   8.00000
mean   13.375000   2.00000
std     3.700869   0.92582
min     9.000000   1.00000
25%    11.250000   1.00000
50%    13.000000   2.00000
75%    15.000000   3.00000
max    20.000000   3.00000
```